UNIVERSAL PROPERTY OF SHEAFIFICATION FOR MAPS INTO SHEAVES OF SETS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A map of sheaves of sets is an isomorphism if and only if it is an isomorphism on all stalks (lemma-points-exactness). invoked at proof lines 1824-1827. Reduces the equality G = G# to stalkwise bijectivity.
- Sheafification does not change stalks: G_x = G#_x (lemma-stalk-sheafification). invoked at proof lines 1827-1830. Provides the stalkwise bijectivity.

Source reference. Original lemma 007T: sources/stacks-sheaves/pinned-original-source.tex, source lines 1558–1558. Cited at proof lines 1824-1827 to reduce G = G# to stalkwise bijectivity; load-bearing inside the counted span.

Source reference. Original lemma 007Z: sources/stacks-sheaves/pinned-original-source.tex, source lines 1779–1779. Cited at proof lines 1827-1830 for the stalkwise bijectivity; load-bearing inside the counted span.

Lemma 0.2. *Let $\mathcal{F}$ be a presheaf of sets on X. Any map $\mathcal{F} \to \mathcal{G}$ into a sheaf of sets factors uniquely as $\mathcal{F} \to \mathcal{F}^\# \to \mathcal{G}$.*

Proof. Clearly, there is a commutative diagram

$$\begin{array}{ccccc} \mathcal{F} & \longrightarrow & \mathcal{F}^\# & \longrightarrow & \Pi(\mathcal{F}) \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{G} & \longrightarrow & \mathcal{G}^\# & \longrightarrow & \Pi(\mathcal{G}) \end{array}$$

So it suffices to prove that $\mathcal{G} = \mathcal{G}^\#$. To see this it suffices to prove, for every point $x \in X$ the map $\mathcal{G}_x \to \mathcal{G}_x^\#$ is bijective, by Lemma 007T. And this is Lemma 007Z above. □

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